Part-I Glass manifiturring

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Glass Preform Manifaturing:

Introduction: manifacturing a glass preform involves several steps that transform the raw materials into a glass preform.

Glass Preform is preform which is used draw the Optical fiber from it.

Raw Material for Preform prepration:-

ⓐ Silica Sand :- (SiO_2)
High-purity silica i.e SiO_2 is primary material.

ⓑ Dopants :- (GeO_2)
This is important for The refractive index of glass preform or fiber maintaining.

But we know that Silica is most Abandent materials on the earth. so this we get in the Form of Sand.

But discussion on this preform making having two Mejor steps along with a utilities ① Core Deposition
② SOC Deposition

Batch Mixing / $SiCl_4$

$SiCl_4$ inthe Form of liquid gas
$GeCl_4$ in the Form of liquid gas.

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① Core Deposition having purity of SiO_2 is very very high 99.9997 pure to make core of optical Fiber.

② SOC Deposition having purity of SiO_2 is less compare to Core Deposition. this used as Clad of optical fiber.

∴ raw material For Core deposition

$SiCl_4$ (g)
$GeCl_4$ (g)
H_2O_2 (g)
O_2 (g)

∴ Row material For SOC deposition

$SiCl_4$
H_2O_2

For melting of glass

Induction Furnace gases.

∴ Argon gas or
Helium gas or

Nitrogen gas

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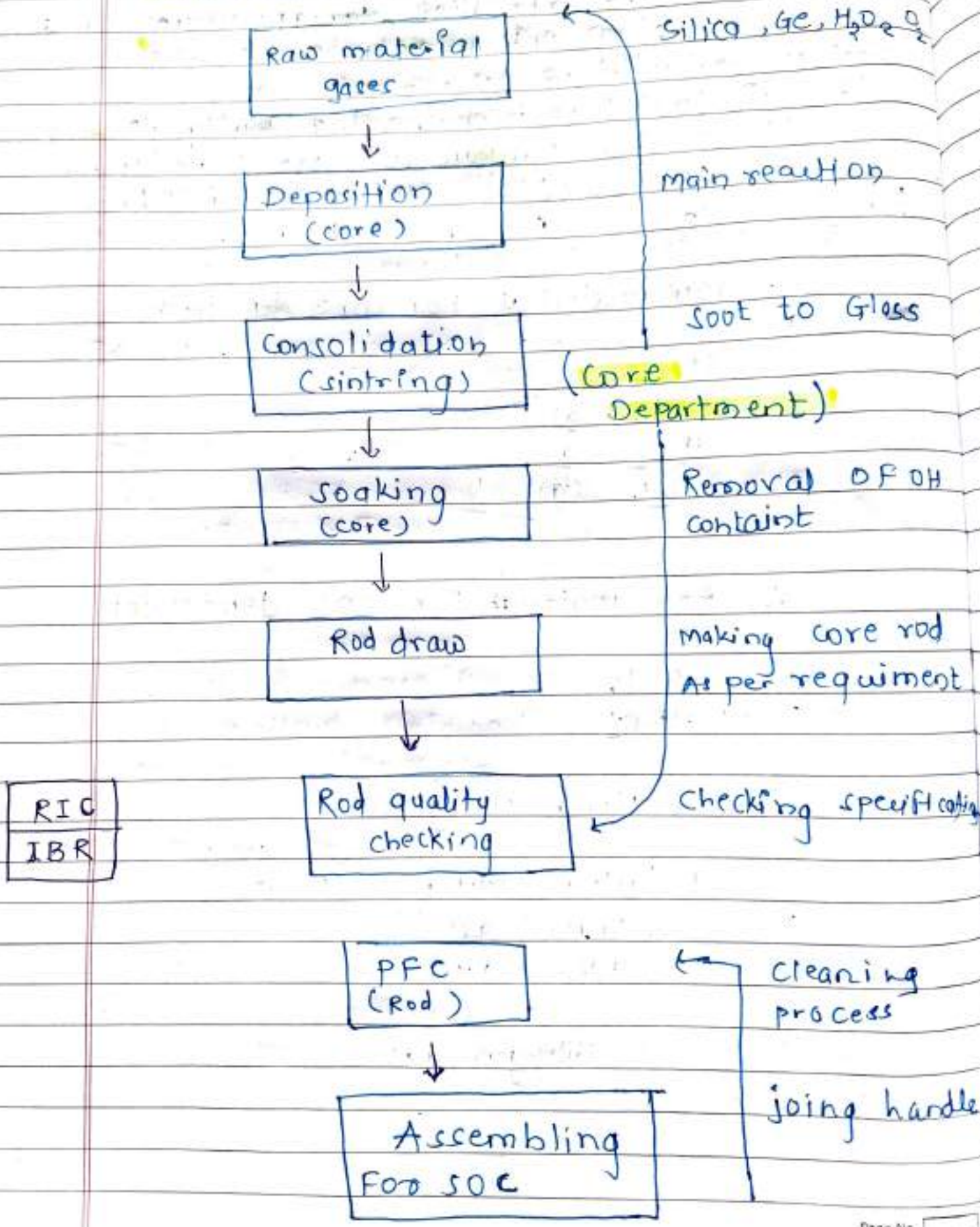
Basic steps of preform manifuring are



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Silica
$SiCl_4 + H_2O_2$

SOC Deposition (Cladding)

↓

Glass formation

SOC Sintring

↓

Removeral of OH molecules

SOC soaking

↓

Checking parameters

IPQ check

↓

coneshaping for draw furnace

coning of preform

↓

To save timeloss in Draw tower

preform joining

↓

Cleaning Outer surface particles

PFC preform

SOC Department

Glass preform

Draw tower

PT

QC

Dispatch

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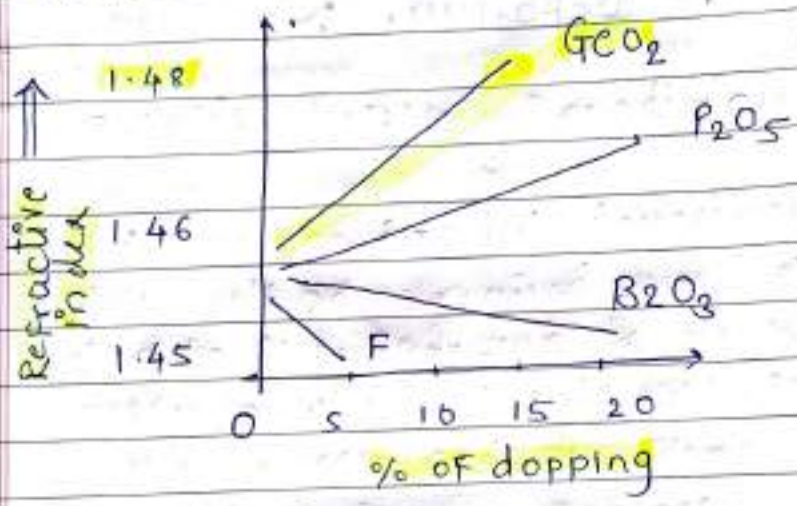
Core Deposition :-

Glass is made up of metal oxide having a "Refractive index" 1.458 at an 850 nm.
For changing the refractive index of glass oxides such as GeO_2 are added as dopants.

Variation of refractive index with doping



∴ GeO_2 increase the refractive index
One important criteria is that the R I of core is greater than that of cladding.

<u>Requirement</u> :-
clear defect free core rod for cladding or soc or Rod in cylinder i.e (RIC)
Because fiber requiment are
① It must be transparent for transmission of light
② R.I core > R.I Cladding
③ Attenuation free rods.

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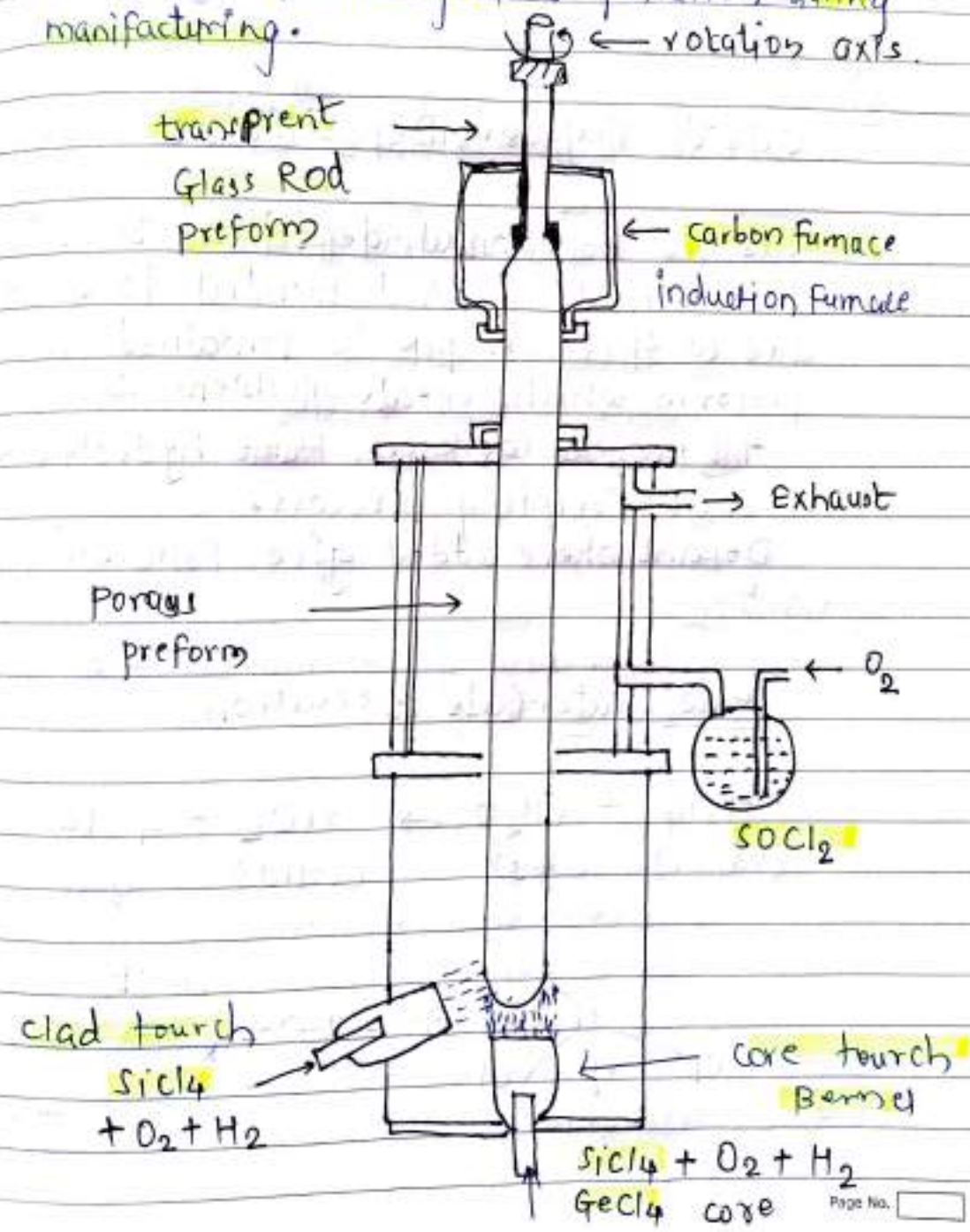
Deposition Methods :-

① VAD (Vapor Axial Deposition) :-

Gases react in Flame to deposite soot on the end of a growing preform, Forming glass in axial direction

This method does not required mendrel.

This method having less pollution during manifacturing.



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The vaporized constituents are injected from burners and react to form a silica soot by flame hydrolysis and make solid porous glass preform.

Dehydrated with $SOCl_2$ using a reaction

$$H_2O + SOCl_2 \xrightarrow{heat} 2HCl + SO_2$$

(vapour) (vapour) (gas) (gas)

(2) Outside Vapour Deposition :- (OVD)

This is horizontal deposition techniques. This method required mendrel to deposition. due to this air gap is remained inside preform which create problem in attenuation.

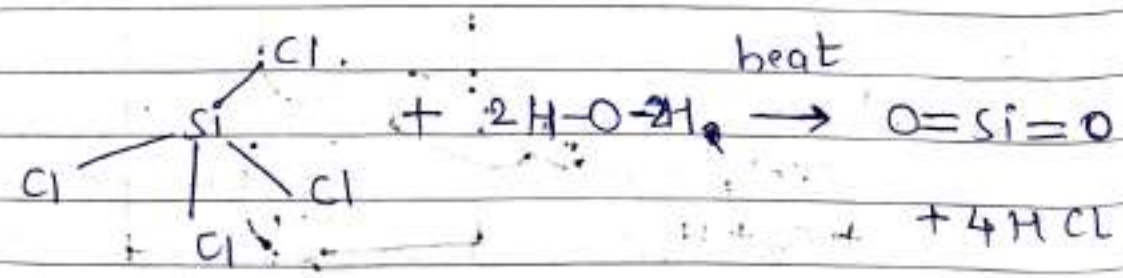
This process work on flame hydrolysis work on soot forming process.

Dopant where added give following Reaction

Raw materials :- Reaction.

$$SiCl_4 + 2H_2O \rightarrow SiO_2 + 4HCl$$

(vapor) (gas) (solid) (gas)



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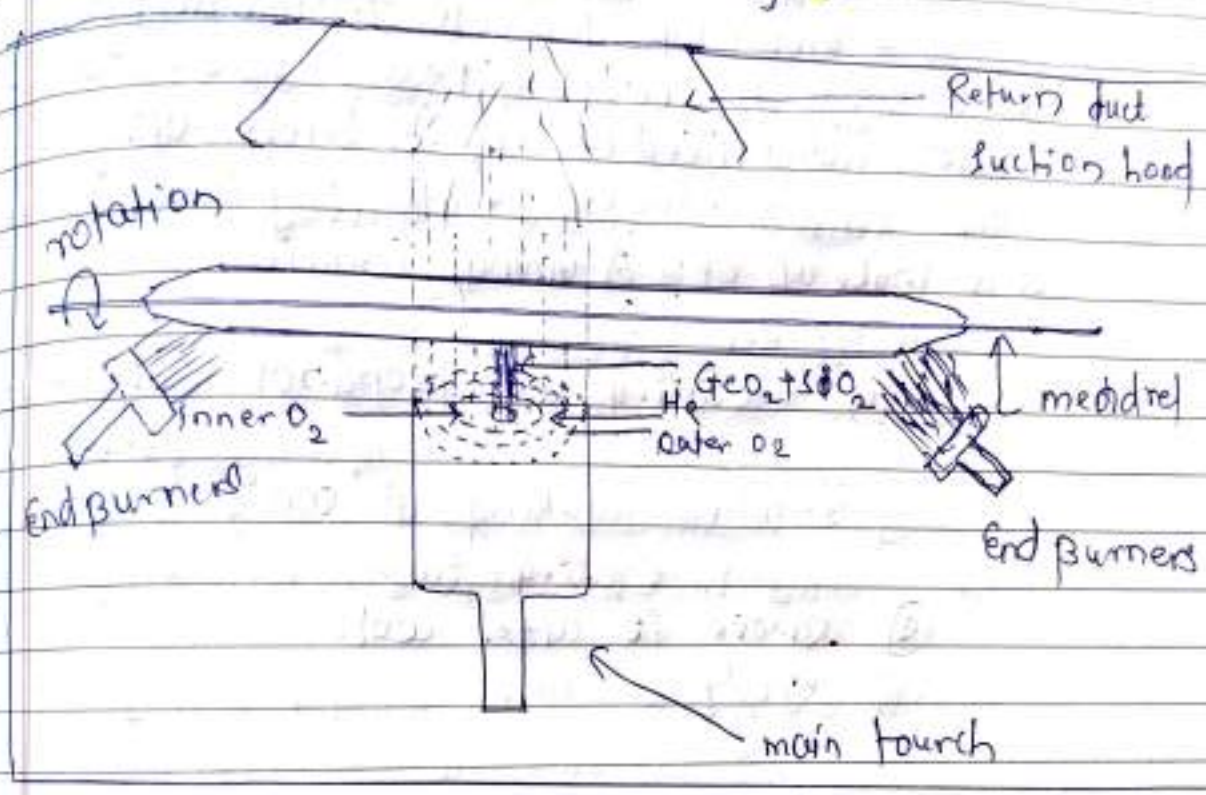
$$\underset{\text{(vapor)}}{SiCl_4} + \underset{\text{(gas)}}{O_2} \rightarrow \underset{\text{(solid)}}{SiO_2} + \underset{\text{(gas)}}{2Cl_2}$$

$$\underset{\text{(vapour)}}{GeCl_4} + \underset{\text{(gas)}}{O_2} \rightarrow \underset{\text{(solid)}}{GeO_2} + \underset{\text{(gas)}}{2Cl_2}$$



$SiO_2 + GeO_2$ = core

inner O_2
H_2
outer O_2

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③ Modified chemical vapor Deposition (MCVD):

④ plasma chemical vapor Deposition

Some important points related to Deposition

① Mendrel:

length = 1600 mm

OD = 8mm Top decreate 7.5 mm at Bottom.

No. of Runs it could be reuse 200-250 preform

New mendrel need to check before using like ID, OD cracks, length, physical apperence.

It is make up of Aluming ceramic

Reuse process

① cleaning with propanol (Remove coot particle)

② 3-4 hrs soaking at 500°C in a soaking tube

③ Remove it after cool.

④ Ready to use.

② Handle:

Inner dia = 12.1 mm to 12.8 mm

Ball OD = 25 mm

Handle = 18 mm

Orings in handle 5 required.

3-Oring in handle to remove mendrel.

2-Oring in chuck to hold mendrel.

O-ring size 2.4 X 7.6

Handle fixing for techniques is check checking check to tip of handle.

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Silica tank / Buffer tank :-

Silica tank required temp which is maintaining 350 to 360°C

N_2 value used to Venting

Ge tank this tank weight near about 344 kg. and actual Ge weight is 92 kg.

preheater is used heating tank temp 60°C

Came line temp should be 100 - 110°C required

The suction of primary duct must required 3000 ± 200 F/min. (hood suction)

The secondary suction must be 320 - 350 Ft/min

Some Important Componant of Deposition

① Main tourch :- It should clean up with ultra sonic after 60 - 62 days.

② End burner :- It should clean up with ultra sonic after 90 - 92 days.

③ Acetalene tourch :- 180 day to be clean

④ V-grove - 120 day to change

⑤ FIC - 15 day - 30 day

⑥ pri, sec - 60 day

⑦ Back Filter - 90 day

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Recipe of Deposition :-

There are total 8 modes in a run

1. Mendrel flame polish mode
2. Carbon coating mode
3. Carbon flame polish mode
4. Centerline mode ($SiCl_4$ only)
5. Core A mode ($SiCl_4$ + $GeCl_4$)
6. Core B mode ($SiCl_4$ + $GeCl_4$)
7. Clad mode ($SiCl_4$ only) OSC1
8. Clad mode ($SiCl_4$ only) OSC2

Manual mode 105 core weight mode

pass based

A, B, C, D, E, F, G, H, I

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① Mendrel Flame polish mode :-

Heating up the mendrel with a flame for two traverse passes.
This passes is to clean off dust particles from the mendrel surface to avoid contamination on center line for carbon coating between mendrel and center line will eventually be burnt off
Note :- To high temp of a mendrel polish flame may Damage the mendrel.

② Carbon coating mode :-

In this mode carbon coating applies on the mendrel before depositing soot. The carbon coating layer act as a lubricant when pulling out the mendrel.
When we facing mendrel removal difficult then increase acetylene flow or number of the carbon coating passes.

③ Carbon Flame polish :-

A few pass of slow flame polish for removing the loose carbon particles and smootheing out the coating surface.

④ Centerline Mode : sicly only :-

Soot deposition starts by depositing sicly only for the centre of the core (Initial soot passes) These centerline passes are of the tools for a indexed proffle turning.
The index of the centerline can be adjusted by controlling the

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Density of centerline.
Since centerline density affect GeCl4 migration to the inside center.

⑤ Core A mode (SiCl4 + GeCl4) :-

After a centerline mode GeCl4 is added into the SiCl4. purpose of raising refractive index.
Amount of GeCl4 used directly the core preform's. index profile thus this is where profile should be tuned.

⑥ Core B mode :- (SiCl4 + GeCl4) :-

Some function as core A mode. the reason for breaking core mode into A and B is to give more flexibility of index profile tuning.

Note :-
Total amount of SiCl4 + GeCl4 during both core A and core B mode is always mentained 40g/min.

⑦ Clad mode :→ Oscillation I (SiCl4 only) :-

Deposition of core preform over clad.
The refractive index of soot deposited during this mode should give an index closed to or equals zero.

⑧ Oscillation 2 :→ Dividing cladding process to two different mode 1, 2 give more flexibilities in tuning density for smoothness.

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Refractive Index :-

Before understanding R.I we need to under stand Nature of light.

Nature of Light :-

" The partical nature arises from the observation that light energy is always emitted or absorbed in the discrite unit called "quanta" or "photons".

The relationship between the energy E and frequency ν of a photon is given by

$$\therefore \quad E = h\nu \quad \text{—— (1)}$$

where h plank's constant 6.625×10^{-34} J.s
$= 4.14$ eV-s

$$\therefore \quad E(ev) = \frac{1.2406}{\lambda\,(\mu m)}$$

we know that

Velocity = Freqn × Wavelength

$$v = f \times \lambda$$

Optical spectrum

Radio	10^{4}
microwave	10^{-2}
IR	10^{-5} → Fibre oprate in IR
UV	10^{-8}
Gamma	10^{-12}

so speed of light in vacuum $c = 3 \times 10^{8}$ m/s

$\boxed{\text{speed}}$ of light in medium $v = 1.4 \times 10^{8}$ m/s.

freqn

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we know that

$$c = v\lambda \quad \text{--- (2)}$$

The ratio of the speed of light in a vaccum to that in matter is the index of refraction n of the material is given by

$$n = \frac{c}{v} \quad \text{--- (3)}$$

$\therefore$ RI $= n$

$\therefore n_g =$ R.I of ~~glass~~ medium

$$n_g = \frac{3 \times 10^8 \text{ m/s}}{1.4 \times 10^8 \text{ m/s}}$$

RI $= 2.14$

But R.I of glass is 1.48

$$\therefore v = \frac{c}{n_g}$$

$$= \frac{3 \times 10^8 \text{ m/s}}{1.48 \text{ m/s}}$$

$\therefore$ speed of light in glass medium is

R.I $= 2 \times 10^8 \text{ m s}^{-1}$

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Reflection Refracted rays Light

when a light ray travels in a two medium or materials or results speed difference of light ray w.r.t materials or medium which is lead to refraction of light ray or the bending of light ray because of different refractive indices.

i.e water 1.33 n_2

Dimond 2.42 n_1

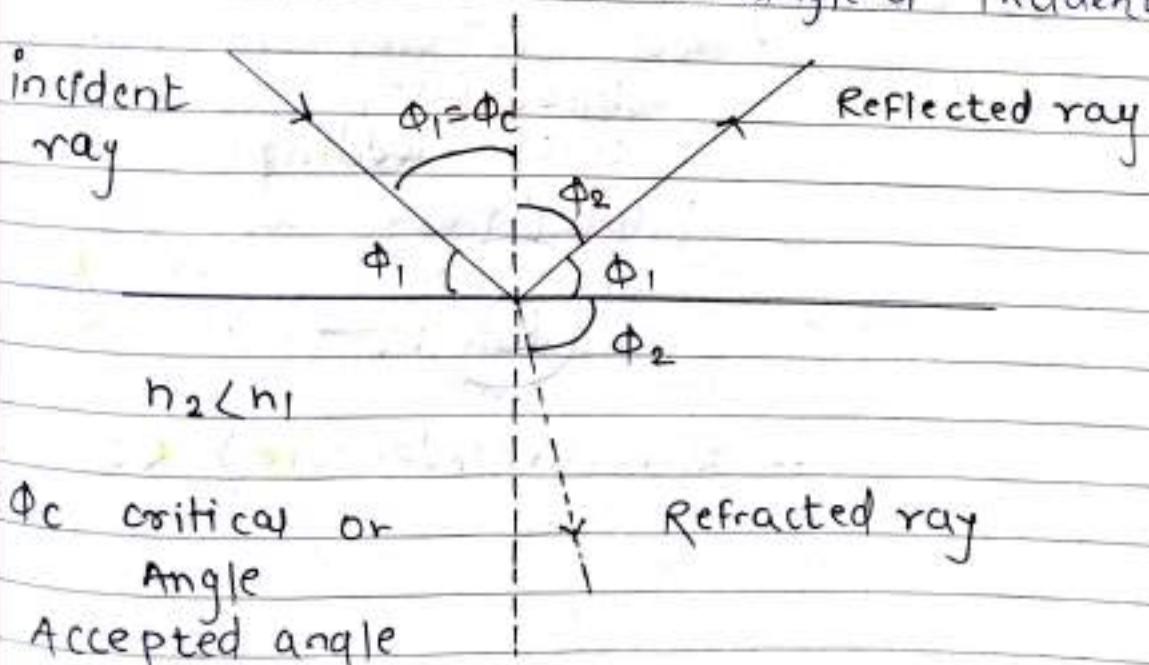
$$n_2 < n_1$$

The relationship at the interference is known as snell's law is given by

$$n_1 \sin\phi_1 = n_2 \sin\phi_2 \quad \text{or}$$

$$n_1 \cos\phi_1 = n_2 \cos\phi_2$$

The angle ϕ_1 between the incident ray and normal to the surface is known as angle of incident



ϕ_c critical or Angle

Accepted angle

$$\sin\phi_c = \frac{n_2}{n_1}$$

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Eg using $n_1 = 1.48$ for glass medium
$n_2 = 1.00$ for air medium

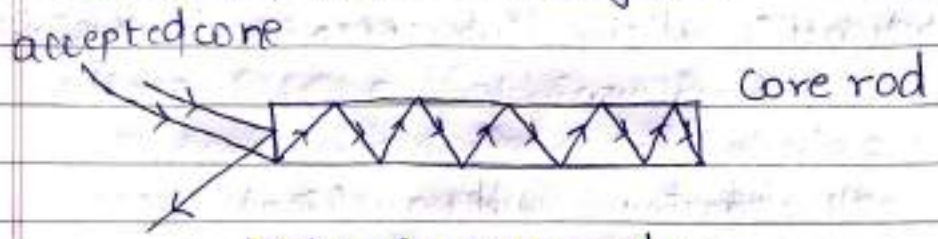
$\sin\phi_c$ = critical angle or accepted angle $= \frac{n_2}{n_1} = \frac{1.00}{1.48}$

$\phi_c = \sin^{-1}$

$\phi_c = 42.5°$

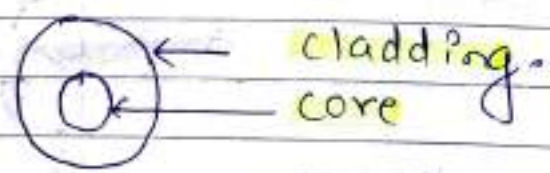
From above

light in glass incident on the interface at angle 42.5° is totally refracted back into glass



Light in core rod or core fiber GeO_2 added to increase refractive index of core compare to cladding silica.

1.48 core
1.458 cladding



∴ Refractive index core > RI cladding

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Due to this phenomenon maximum part of signal travels through core region.

Core :-

Core is the heart of the optical fibre and it governs the important optical parameter like CUTOFF, MFD, ZD, Attenuation

Core is the area where 99 % of signal travel through it

core / clad is ratio of D/d its very important parameter to controlling all critical parameter of core.

Higher is D/d value always give the good optical Result.

Centerline refractive index :-

The refractive index of the center of the core preform can be tuned by controlling the density of these passes.

During sintering $GeCl_4$ deposited at core mode will diffuse into the centerline.

The level of migration is related closely to centerline density. Lower centerline density make diffusion easier and can rise the refractive index.

It can be tuned by, speed, $SiCl_4$ flow, and number of passes.

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Refractive index Delta :-

Ge migration/diffusion into the centerline and outside to the clad during drying and sintering stages, slopes in profile indicate migration.

Refractive index delta plot is mostly determined by deposition of $GeCl_4$ which take placed during core mode A and core mode B.

The perpose of two modes is mostly to allow more flexibility in tuning refractive index profile.

$GeCl_4$ and $SiCl_4$ flow according to allow exponential equations. From the recipe core mode A and core mode B have their own flow ramping equation. This enables the $GeCl_4$ & $SiCl_4$ flows to ramp down, up or stay constant in both mode.

Note :- ① All gas flow except $GeCl_4$ & $SiCl_4$ flow have only one ramping eqn through out core mode A, core mode B

② Instruction how to set ramping eqn can be found in recipe command spread sheet.

eg ① Profile with high dip

① In this profile more Gecly can be added at the beginning of core A mode, so that more can be deposited into the centerline an brings centerline index up.

② To lower the peak's, less Gecly flow that at the core A mode which is also the begin of core B mode since the peaks occurse at about the middle of core deposition which is closed to core A and core B transition according recipes. As a result Gecly should be ram down during core A start higher and end lower flow.

eg ② Delta with slanted slope :-

Slant :- To incline away from the a relatively straight surface or lined used as refrence.

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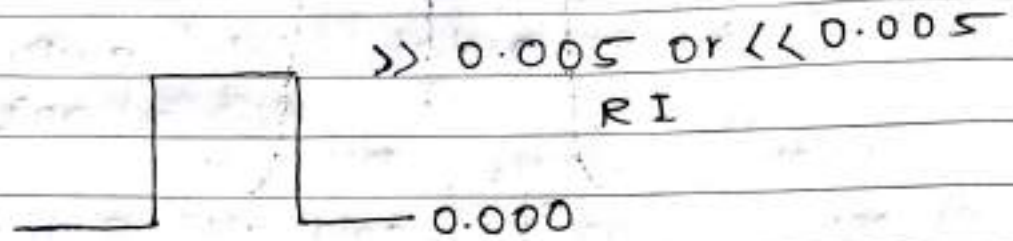
This can be more squred, by increasing GeCl4 at end of core B which correspond to the edges of the delta and increase the indexes of the edges

Thus set ramping up equation for GeCl4 flows during core B mode will be help full.

Eq ③ **Delta height Adjustment :-**



If delta is too high or low a way to tune is kept the GeCl4 flow pattern during core deposition mode (A, B) and increase and the decrease flows by a percentage similar to the index deviation percentage.

Changing SiCl4 and GeCl4 Flows in Recipe

The total amount of silica and GeCl4 used during core mode A and B is always 45 g/min
22.5 g/min/torch

∴ In order to maintain the sum, when increase GeCl4 flow, silica flow has to be decreased and vice versa.

When program switches from core B mode to clad mode, each SiCl4 MFC should have flow rate of 11.25 g/min

Eg if GeCl4 flow has to be increased from 1.8 g/min to 2 g/min per MFC.

Before :-

$$2 \times \text{GeCl}_4 \text{ MFCs} = 2 \times 1.8 \text{ g/min}$$
$$= 3.6 \text{ g/min}$$

$$4 \times \text{SiCl}_4 \text{ MFCs} = 4 \times 10.35 \text{ g/min}$$
$$= 41.4 \text{ g/min}$$

$$\therefore \text{Total Flow} = 3.6 + 41.4$$
$$= 45 \text{ g/min}$$

After :-

$$2 \times \text{GeCl}_4 \text{ MFCs} = 2 \times 2.0 \text{ g/min} = 4$$
$$4 \times \text{SiCl}_4 \text{ MFCs} = 4 \times 10.25 \text{ g/min} = 41$$
$$\therefore \text{Total Flow} = 4 + 41$$
$$= 45 \text{ g/min}$$

However the factor should be -0.5 times of GeCl4 This is because SiCl4 use 4 MFC while GeCl4 has only 2 and -ve sign is compensate for increasing decreasing GeCl4.

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Production challenge or problems in core deposition

① **Ripples :-** It is nothing but rings formation on soot preforms.

Following action to be take on Ripples.

① lower traverse speed but it will also decrease deposition rate and increase density.

② changing total interlace OFFset in the recipe

③ setting a Flip-Flop function changes fast and slow traverse direction every number of passes

④ Vibration in soot

② **Density control :-**

Low density causing soot cracking preform become too big

Density to be increase points are

① lower traverse speed
② Increase hytrogen (heat)
③ Decrease silica flow rate
④ Increase makeup O_2

③ **Deposition Rate :-**

① Faster traverse speed
② Hydrogen flow lower
③ lower density
④ lower SiCly flow
⑤ Fume tube

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④ **Diameter variation** :

① Alignment
② Exhaust Flow
③ intake air (filters)
④ mendrel condition

Otherwise check sintering temp, speed

⑤ **Splits** :-

① Low density of soot
② End burner heat
③ Alignment burners
④ mostly end burners heat and low density.

⑥ **Warts**

lower makeup Oxygen Flow

⑦ **Bubbles**

① mostly related to centerline
② mendrel removal process
③ soaking time, temp.
④ check vaccum in sintering
⑤ Bottom plug
⑥ Rod draw vaccum
⑦ Rod draw temp
⑧ Rod draw speed

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Core Glass steps

Mendrel prepration

↓

core deposition

↓

purging storage of soot — soot storage with N_2

↓

Mendrel Removal — pulling mendrel 4 psi (fixing bottom plug).

↓

Sintring — (soot to glass formation)

↓

Soaking — (Soaking with N_2)

↓

Rod draw — (Rod draw of core preform cut rod as per lenght, dia)

↓

cutting proffling — (end cutting profile of rod)

↓

IPQA — checking all parameter.

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① Mendrel Prepration :-

Mendrel is rod made up of Alumirum rod having ~~diag~~ dia 8mm TOP - 7.5 mm at bottom.

core deposition reaction take place on to this mendrel inside deposition.

This mendrel is reuse for deposition upto the 200-250 preform deposition.

But before reuse it has process of cleaning and prepration for use.

cleaning

① scrubbing along length to remove soot particles

② Cleaning with ~~Sa~~ ~~ak~~ durex paper with propanol.

③ put this into soaking at 500°c temp in an soaking tube, for 3-4 hrs

④ Remove it after soaking cool it at room temp.

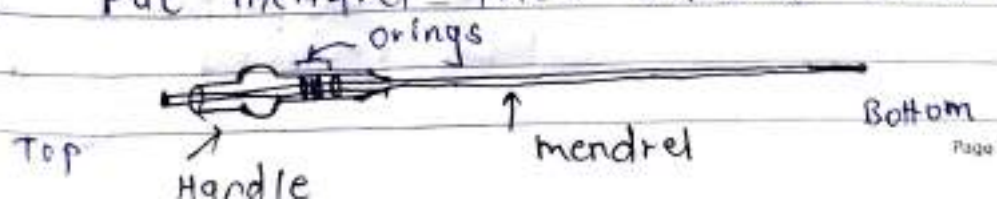
prepration

① Take clean mendrel,

② take proper handle having ID: 12.1 mm
BallOD = 25 mm
Handle OD = 18 mm

③ take 5-orings with proper size 2.4 x 7.6

④ put mendrel into handle with orings.



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② **Core deposition** :-

This is horizontal deposition techniques where deposition reaction take place on mendrel.

$SiCl_4 + 2H_2O \rightarrow SiO_2 + 4HCl \uparrow$

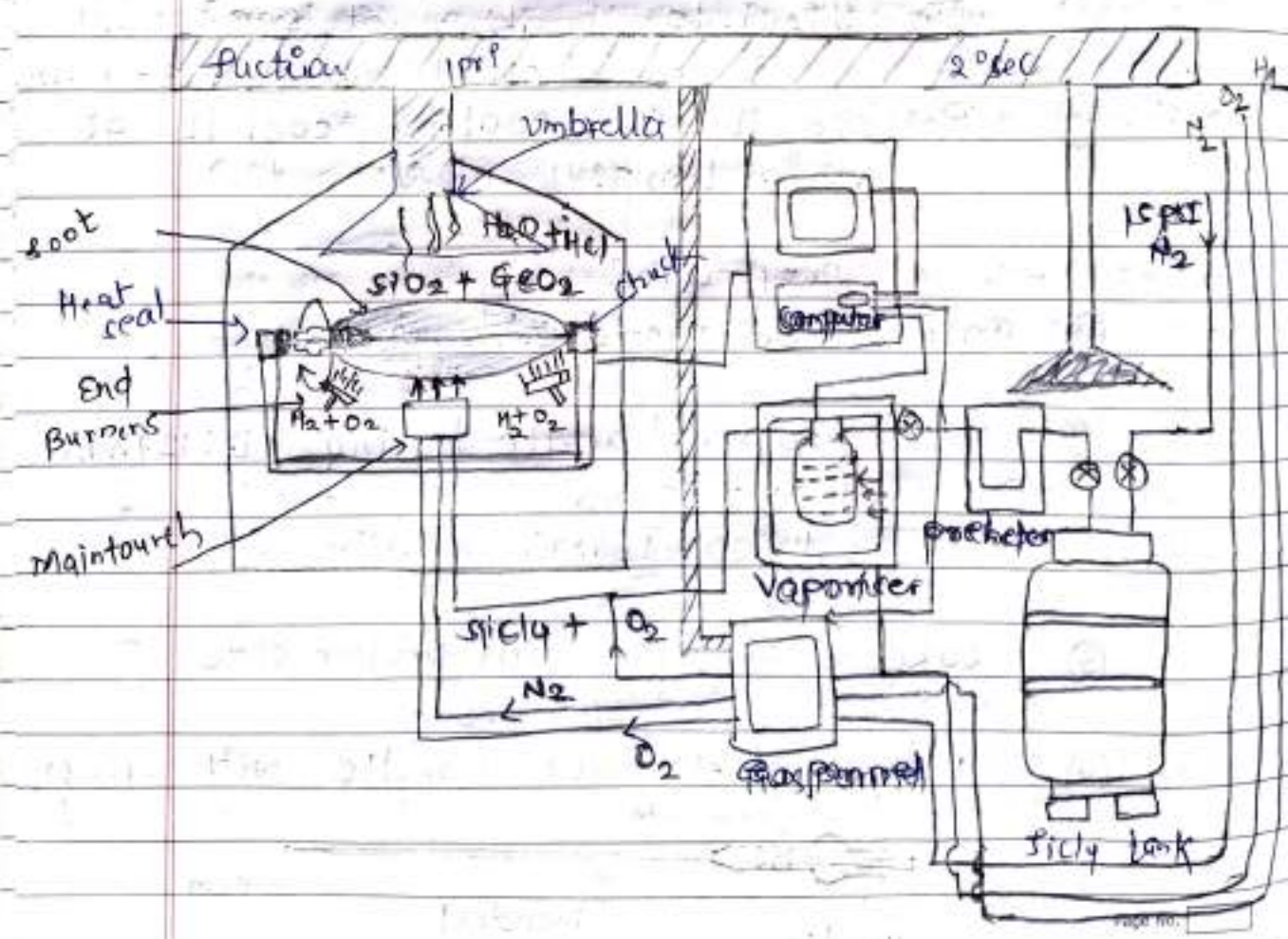
$SiCl_4 + O_2 \rightarrow SiO_2 + 2Cl_2$

$GeCl_4 + O_2 \rightarrow GeO_2 + 2Cl_2$

Deposition consist of following main compont

Main tourch

End burners



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① Gecly tank : storage of dopant in a tank

② Sicly tank : storage of silica

③ Vaporiser :- It create vapour from liquid silica

④ preheater :- Silica line must be having temp so it is used to heating continious to the silica

⑤ computer :- It consist of all PLC and Gas control software where can set the recipe.

⑥ Gas pannel :- It consist of all MFC with repective gases like silica, Ge, H_2, O_2, N_2 etc.

⑦ Traverse bed :- It consist of motor for traversing soot because in OVD torches are fixed one at the centre.

⑦ Heat seal (chuck) :- It hold both end of soot or mendrel and keeping rotating soot.

⑧ Hood (Umbrella) :- It is used to collect the polution of H_2O + HCl through suction.

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Some important Note related to process.

1. mendrel flame polish mode heating up the 2 traverse passes.

2. Carbon coating mode applies carbon coating to the mendrel before depositing soot. It act as lubricant. when pulling out mendrel.

3. In order to avoid problems with closing the hole in the center of core preform only SiCl4 is deposited for initial soot passes.

4. The refractive index of core preform can be tuned by controlling the density of these centerline passes.
   - Lower density can raise the refractive index.

5. We use GeCl4 insted of silica (initially) it will affects the core profile by increasing the delta.

6. Higher concentration of GeCl4 in SiCl4 increase the scattering loss in the final o/p. (fibre)

7. It is observed that
   if no. of passes is decreased in centre line mode of SiCl4 High of dip will be reduced rapidly.

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(5) The amount of Gecly and sicly is deposited on the mendrel. at manual mode A.

(9) The amount of Gecly that get deposited in stage of process defines the peak of the profile.

(10) The initial density of soot plays a major role in preventing the split of the soot.
pass 30th to 107th pass

(11) core mode A region where ramping of Gecly take place.
This region perticulary defines a slope of the shape which is called "alpha"

Ramping Eqn followed by both Gecly is

$$Y = m X^n + C$$

where

m = factor
X = no. of passes
n = exponant
C = initial flow
Y = flow at Xth pass.

all flow other than Ge (Si, H_2, O_2) remains constant.

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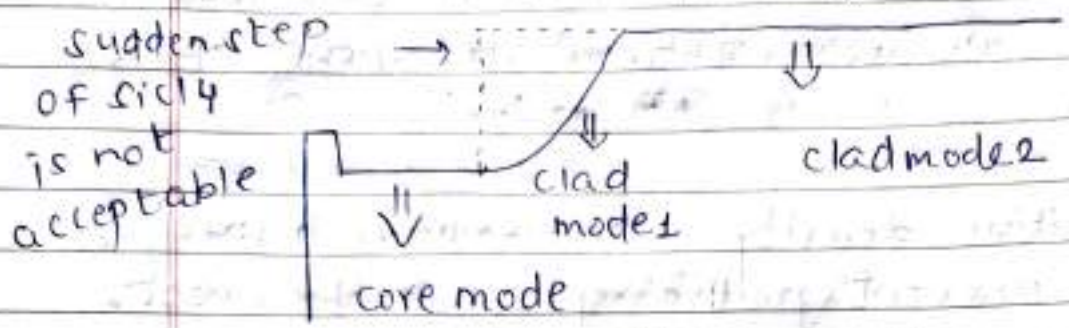
(12) The reson for breaking core mode into A and B is to give more flexibility of index profile tuning.



(13) Here we do density matching by ramping Hydrogen from end of core mode to clad 2 mode. If not done like this high poribility of soot slippage.

(14) The nature of the transition depends on the nature of clad 1 mode.

(15) Density play major role

Profile Parameter

1) D/d :- "This is ratio of clad mode to core mode"

If the D/d is very less, the core region is very close to the atmosphere hence very sensitive to OH "

D/d is the lone factor to decide SOC to be deposited

product

G652D — 2504 — Delta >= 3.55

G652D — 2004 — >= 3.55

G657 — 2.5 - 2.6

2) Delta :- "Difference between the refractive index of the core and clad is called delta"

charectristics of Higher delta and Lower delta it impact on Optical properties.

Higher delta

1. Higher delta gives good micro bending performance.
2. also gives the lower MFD
3. also affect lower ZD
4. Higher cutoff & Attenuation.

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Lower Delta

1. High macro bending performance.
2. Higher MFD
3. Higher ZD
4. Lower cutoff
5. Lower Attenuation

so we chose an optimum delta for a waveguide performance per different product

G652D → 250 μ 4.6 - 5.1
→ 200 μ 5.1 - 5.6
→ 250 μ 5.1 - 5.6

G657 — 4.9 - 5.1

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(3) Alpha :-

" The variation of the refractive index as a function of radius till core boundary is defined by Equation ".

$$n(r) = n_1 \sqrt{1 - 2\Delta (r/d)^{\alpha}} \quad r \leq \alpha$$

where

$$\Delta = \frac{n_1^2 - n_2^2}{2n_1^2}$$

specs For alpha > 6

High alpha Advantages

- Alpha is factor to control the waveguide dispersion.
- Alpha is higher waveguide dispersion will be lower vice versa.
- But scattring losses is more with higher alpha.

(4) Centerline width

(5) Centerline height

(6) Rod diameter

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Profile problem :-

1. Negative shape :- (MFD)
2. tip of dip or Central width (
3. Lower : alpha (waveguide)
4. Low/high Delta (waveguide)
5. Low/high D/d ( )
6. Delta with slanted ( )

Waveguide parameter

1. MFD (mode field diameter) > core dia (9.1-9.3)
2. CutOFF 1160 - 1360 nm
3. zero dispersion (ZD) 1304 - 1324 nm
4. Cromatic dispersion (CD) > 0.17,

Attenuation's parameters

1. Att^n - 1310 nm 0.34 dB/km
2. Att^n - 1550 nm 0.21 or 0.24
3. max - 1625 nm 0.
4. water peak — 1383 nm 0.34 or 0.31

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(3) **Sintering :-**

"Sintering is important process of glass manifacturing after deposition of soot it get converted into pure glass at sinter stage along with removal of OH molecules from soot with precence of a or with the help chloride, or Florine, at specific temp at 1500 - 1550 °C"

There is specific zones of temp

1400 - 1450

1450 - 1500

1500 - 1550

1550 - 1600

Sinter consist of heating Element and Glass tube high temprature sensitive

(4) **Soaking :-**

Sinter preform get soak into soaking at lower temp for some perticular time to remove trap gases.

(5) **Rod draw :-**

Soaked preform goes to rod draw to draw desired diameter with proper length.

(6) **Profile & cutting :-** Checking profile cutting as per SOC requirement or RIC requirement

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⑦ **PFC :-** polly Flow control (HF cleaning)

PFC is area where rod or preform surface cleaning with the help of HF as strong acid.

HF Remove or we can say etching the surface of glass and Remove surface particles or rod or preform.

Cladding part

⑧ **Rod Assembly prepration :-**

joining both end with scrap handle provide to PFC again to ready for the SOC deposition.

⑨ **SOC or Clad Deposition :-**

Clad Deposition is the area where the clad is deposited on surface of the core rod as per CSA and D/d along with preform diameter requirement

Important parameter of SOC

1) CSA of rod
2) D/d of rod
3) preform diameter of draw Tower.

Here again

$$SiCl_4 + H_2O_2 \rightarrow SiO_2 + HCl$$

But purity of SiCl4 less compare to core Deposition

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(10) **clad sintering :**

Clad sintering convert the soc preform soot into transparent Glass preform which used as Final product of Glass preform used to draw at draw tower. Here is measure steps involment of He gas to convert soot to clear glass using temp near 1500°c, rest same as core sin

(11) **clad soaking :-** preform soaking at perticular temp.

(12) **preform conning :-** Induction furnace is used to coning at one end to avoid cutOFF Failure at rampup in draw tower.

(13) **preform joining :-** To avid time loss in draw tower the preform will be joined dubble, tripple etc.

(14) **preform pfc :** Before sending draw quality must checked and to remove surface particle again HF cleaning done.

(15) **Quality check** cheaking all parameter of preform send to draw tower.

"We will see part - 2 more details **Fiber manifacturing**"

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Fiber manifacturing "